Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Yongxin Chen	Email	yongchen@gatech.edu
Institute Name	Georgia Institute of Technology			
Address	CODA E1048B, 756 W Peachtree St NW, Atlanta, GA 30308			

2. Technology Overview

Title of Proposed Technology	Multimodal Large Language Model-Powered Agentic AI System for Enhanced Clinical Diagnosis			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	MRI	*Therapeutic Devices & Diagnostic Devices*	X-ray and CT
Concept of Proposed Technology	Diagnostic errors affect an estimated **10–15%** of medical cases globally and contribute to substantial preventable harm. In many situations, errors arise not from missing information but from the **cognitive burden of integrating complex, multimodal clinical data** (imaging, labs, vitals, prior notes, waveforms, and longitudinal history) under time constraints. This project proposes an **agentic AI clinical decision-support system** powered by **multimodal large language models (MLLMs)**. The system will function as a diagnostic partner to clinicians by synthesizing multimodal patient data for **differential diagnosis and clinical QA**, recommending **next-step tests** and workups aligned with clinical guidelines and uncertainty reduction, and			

(max 10 pages)

	producing **trustworthy, auditable draft reports** (e.g., radiology-style summaries) with supporting evidence. Key features: • **MLLM core** capable of jointly processing heterogeneous clinical inputs (e.g., X-ray/CT/MRI, structured EHR features, clinical text, and waveforms such as ECG/EEG where available) to support tasks including differential diagnosis, question answering, and summarization. • **Agentic system layer** that orchestrates tool use and agentic skills (e.g., guideline lookup, retrieval from hospital knowledge base, longitudinal timeline construction), enabling interactive collaboration with clinicians for diagnosis, test planning, and report generation. • **Data curation and continual improvement pipeline** combining expert review with LLM-assisted annotation to create high-quality, task-specific training and evaluation data. • **Post-training methods tailored to agentic clinical workflows**, improving accuracy, reliability, and adherence to safety constraints (e.g., preference tuning, instruction tuning, and reinforcement learning).
Rationale for Joint R&D Project	Translating MLLM-based agentic technologies into a clinically usable diagnostic assistant requires both **frontier AI capability** and **deep integration with real clinical environments**. A joint R&D framework is essential to align model development with clinical workflow constraints, data governance, regulatory requirements, and usability expectations. Rationale for collaboration: • **Research institute strengths:** generative AI, multimodal foundation models, agentic AI architectures, training and inference optimization and evaluation methodology. • **Korean SME partner strengths:** access to real-world clinical data and workflows, clinical UI/UX development, system integration with hospital IT (e.g., PACS/EHR), operational deployment capability, and clinical evaluation coordination. • **Need for iterative co-development:** continuous clinician-in-the-loop feedback is required to meet cost, latency, usability, and safety requirements and to ensure the system supports clinical practice. Expected benefits of joint R&D: • Faster transition from research prototypes to deployable clinical-grade systems. • Reduced technical and operational risk through iterative validation in realistic settings. • Establishment of a scalable data/feedback pipeline enabling continuous performance improvement and product iteration.
Key Words	1. Multimodal large language models (MLLMs) 2. Agentic AI systems 3. Clinical decision support and diagnosis 4. Human-AI collaboration 5. Trustworthy AI for healthcare

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	The project objective is to develop an **agentic MLLM-based diagnostic assistant** that improves clinical decision-making by helping clinicians: (i) reduce diagnostic errors, (ii) plan efficient and appropriate next-step testing, and (iii) generate high-quality clinical reports. Specific R&D objectives: • **Diagnostic assistance:** develop an AI agent that improves accuracy and efficiency of differential diagnosis across multimodal inputs. • **Test planning and workflow support:** develop capabilities to propose clinically plausible next tests/workups, justify recommendations, and streamline report drafting. • **Continuous improvement:** implement a robust **data curation pipeline** and **post-training strategy** enabling traceable and iterative model improvement. • **Clinical validation:** deploy and evaluate prototypes in realistic clinical environments as a key step toward commercialization.			
R&D Target Specifications	Target specifications will be refined jointly with clinical stakeholders during Year 1; initial targets include: Diagnostic performance (multimodal): • **Input modalities:** at minimum MRI and common CT/X-ray series; expandable to structured EHR and waveform data (e.g., ECG/EEG) depending on partner availability and governance. • **Outputs:** ranked differential diagnosis, supporting rationale, and evidence references (e.g., highlighted image regions when applicable, cited prior notes/labs when used). • **Benchmarking:** achieve competitive or state-of-the-art performance on relevant public benchmarks (e.g., medical multimodal QA and radiology reasoning benchmarks) and partner-defined internal test sets. Test planning and report generation: • **Test planning:** propose next-step tests/workups that are clinically plausible, uncertainty-reducing, and consistent with guidelines and patient context; provide justification and alternatives. • **Report generation:** produce structured drafts (e.g., findings) that outperform baseline report-generation systems in factuality, completeness, and clinician preference. Assistant usability and safety in human-in-the-loop use: • **Human factors:** interactive, low-friction UI; supports clinician oversight and editing; provides clear uncertainty signaling.			

(max 10 pages)

	• **Trustworthiness:** provenance-aware outputs (what data was used), refusal behavior where appropriate, and logging for auditability. • **Evaluation:** clinician-rated usefulness and safety in task-based studies; measurable time reduction for routine documentation/report drafting without degrading clinical quality.
R&D Plan	A phased plan emphasizes early feasibility, followed by data and model improvement, then real-world validation. Year 1: System design and feasibility prototypes (MLLM + agent scaffold) • Evaluate zero-shot and few-shot capability of candidate MLLMs on key modalities (X-ray/CT/MRI) for differential diagnosis, QA, and summarization. • Build a **minimum viable agentic system** that collaborates with clinicians for core tasks (diagnosis support, question answering, draft reporting). • Implement tools/skills (e.g., retrieval from curated clinical references, structured data parsing, uncertainty estimation, evidence linking). • Identify failure modes and workflow pain points; define **data requirements** and the **curation strategy** accordingly. Year 2: Data curation pipeline + post-training • Implement the curation pipeline combining: clinician feedback, targeted expert annotation, and LLM-assisted labeling where appropriate. • Deploy small-scale data collection in operational settings (pilot units) with clear governance and de-identification practices. • Develop post-training methods using curated internal data plus high-quality open datasets to improve: diagnostic accuracy, evidence grounding, and interaction behaviors. • Iteratively refine the data pipeline based on measured model weaknesses. Year 3: Clinical evaluation, refinement, and commercialization readiness • Deploy the system in realistic clinical environments (pilot deployment), integrated with partner workflows. • Conduct systematic evaluation versus existing tools and standard practice: diagnostic assistance quality, efficiency gains, report quality, and user trust. • Refine UI/UX, agent policies, and safety guardrails based on clinician feedback and observed errors. • Assess commercialization path (product requirements, cost targets, integration checklist) and prepare for technology transfer.

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will lead core AI research and prototype development, with responsibility for model architecture, training, evaluation, and safety methodology. Key responsibilities: ● Design and implement the **agentic MLLM system** for multimodal clinical reasoning, interaction, and tool orchestration. ● Develop the **data curation pipeline** (schema, labeling protocols, quality control, and evaluation sets). ● Develop and validate **post-training algorithms** tailored to agentic clinical workflows (e.g., preference optimization, reinforcement learning). ● Provide technical leadership to support partner integration, performance evaluation design, and iterative improvement cycles.
Expected R&R of Korean SME Partner	The Korean SME partner will lead clinical data operations, software integration, deployment, and clinical evaluation, enabling translation into a product-grade system. Key responsibilities: ● Integrate the agentic AI system into clinical software ecosystems (e.g., PACS/EHR interfaces, user authentication, audit logging). ● Execute data collection and pipeline operations in compliance with applicable governance, including coordination with clinicians and data custodians. ● Lead productization activities: UI/UX implementation, reliability engineering, latency/cost optimization, and operational monitoring. ● Deploy and evaluate the system in clinical environments, coordinating user studies and iterative feedback for refinement and commercialization planning.

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technical outcomes • A deployable agentic multimodal diagnostic assistant supporting differential diagnosis, test planning, and clinical report drafting with evidence-aware outputs. • A scalable data curation + post-training pipeline enabling continuous system improvement through product iterations. • Validated evaluation methodology and performance results on public benchmarks and partner-defined clinical tasks. Clinical and societal impact • Improved diagnostic accuracy and efficiency through multimodal synthesis and workflow-aligned decision support. • Reduction of preventable harm associated with diagnostic errors by providing consistent reasoning, uncertainty signaling, and structured evidence presentation. • A practical demonstration of human–AI teaming in high-stakes clinical decision-making. Industrial impact • Transfer of a clinically integrated AI capability to the Korean SME partner, including know-how for model deployment, monitoring, and iterative improvement. • Foundation for commercialization of next-generation AI clinical assistants and expansion into adjacent healthcare workflows. • Strengthened competitiveness of Korean SMEs in medical AI and acceleration of safe adoption of AI technologies in healthcare.

6. References

(1) Q. Zhang, **Y. Chen**, “Fast Sampling of Diffusion Models with Exponential Integrator,” International Conference on Learning Representations (ICLR), 2023.

(2) Y. Zhu, W. Guo, J. Choi, P. Molodyk, B. Yuan, M. Tao, and **Y. Chen**, “Enhancing Reasoning for Diffusion LLMs via Distribution Matching Policy Optimization,” preprint, 2025.

(3) B. Yuan, Z. Zhao, P. Molodyk, B. Hu, and **Y. Chen**, “Clarify Before You Draw: Proactive Agents for Robust Text-to-CAD Generation,” preprint, 2026.

(4) S. Li, Y. Zhu, J. Gu, K. Liu, Z. Lin, **Y. Chen**, M. Tao, A. Grover, and J. Kuen, “Lavida-R1: Advancing Reasoning for Unified Multimodal Diffusion Language Models,” preprint, 2026.

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization